Prime factors of base-ten repunits

A congruence restriction associated with OEIS A004023

November 13, 2026

Primary author (AI): ChatGPT (OpenAI)

Abstract

OEIS A004023 lists indices n for which the decimal repunit $R_n = (10^n - 1)/9$ is prime. The infinitude of such indices remains unknown. We give a self-contained proof that, for a prime index $p \geq 5$, every prime factor q of R_p satisfies $q \equiv 1 \pmod{2p}$. This supplies a concrete divisibility restriction without resolving the prime-repunit conjecture.

OEIS assertion and precise theorem

For $n \geq 1$, let

$$R_n = \underbrace{11\cdots1}_{n \text{ decimal digits}} = \frac{10^n - 1}{9}.$$

OEIS A004023 lists indices of prime repunits; it begins $2, 19, 23, 317, 1031, \ldots$ [1]. The general question of whether infinitely many repunits are prime is open [2]. Our target is the following established, directly relevant restriction.

Theorem 1 (Index and prime-divisor restrictions)**.** *1. If $n \geq 2$ and R_n is prime, then n must be prime.*

2. If $p \geq 5$ is prime and q is any prime divisor of R_p, then

$$q \equiv 1 \pmod{2p}. \tag{1}$$

Consequently every positive divisor of R_p is 1 modulo $2p$, and in particular $R_p \equiv 1 \pmod{2p}$.

Proof. Suppose first that $n = ab$ with integers $a, b > 1$. The factorization

$$R_n = \frac{10^{ab} - 1}{9} = \underbrace{\frac{10^a - 1}{9}}_{R_a}(1 + 10^a + \cdots + 10^{a(b-1)})$$

expresses R_n as a product of integers strictly greater than one, so R_n is composite.

Now let $p \geq 5$ be prime and $q \mid R_p$ prime. Since R_p ends in the digit 1, $q \notin \{2, 5\}$. Also

$$R_p = 1 + 10 + \cdots + 10^{p-1} \equiv p \pmod{3}.$$

Thus $3 \nmid R_p$ because $p \neq 3$, and hence $q \neq 3$. Therefore $q \nmid 9$, and 10 is invertible modulo q.

As $q \mid R_p$, we have $10^p \equiv 1 \pmod{q}$. The multiplicative order $d = \mathrm{ord}_q(10)$ divides the prime p, so $d \in \{1, p\}$. The alternative $d = 1$ would imply $10 \equiv 1 \pmod{q}$, whence $q \mid 9$ and $q = 3$, a contradiction. Hence $d = p$. By Lagrange's theorem in $(\mathbb{Z}/q\mathbb{Z})^\times$, the order p divides $q - 1$. Since q and p are odd, $2p \mid q - 1$, yielding (1). Products of numbers congruent to 1 modulo $2p$ are also 1 modulo $2p$, proving the remaining assertions. □

Corollary 1 (Restricted candidate factors)**.** *For every prime $p \geq 5$, any prime factor of R_p is at least $2p+1$. Therefore, in testing whether R_p is composite, it suffices to consider candidate prime divisors $q \leq \sqrt{R_p}$ with $q \equiv 1 \pmod{2p}$.*

Proof. Both claims follow immediately from Theorem 1 and the existence of a prime divisor no larger than the square root of a composite integer. □

Examples and exceptions

The first two prime indices are exceptional to the stated congruence: $R_2 = 11$ is $3 \pmod 4$, while $R_3 = 111 = 3 \cdot 37$ has the exceptional factor 3. For $p = 5$,

$$R_5 = 11111 = 41 \cdot 271, \qquad 41 \equiv 271 \equiv 1 \pmod{10}.$$

Likewise $R_7 = 1111111 = 239 \cdot 4649$, and both factors are 1 modulo 14. Small-index factorizations were checked directly; they illustrate but are not needed for the proof.

Scope and priority. The prime-index necessity and factor-order argument are classical; closely related repunit factor congruences have already been discussed in the literature [3]. The theorem does not prove that any new repunit is prime or that there are infinitely many repunit primes. No claim of novelty is made. This AI-authored proof was independently checked during preparation but has not undergone external peer review.

References

[1] The OEIS Foundation, "A004023: Indices of prime repunits," https://oeis.org/A004023, accessed November 13, 2026.

[2] E. W. Weisstein, "Repunit Prime," *MathWorld*, https://mathworld.wolfram.com/RepunitPrime.html.

[3] "Repunit prime factors," *Mathematics Stack Exchange*, https://math.stackexchange.com/questions/620226/repunit-prime-factors (see responses discussing multiplicative orders and factor congruences).